Modeling progress:
Causal models and the imperfective paradox

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Causal reasoning and causal language

‘Practical’ causal intuitions vs. linguistic causation

- causal reasoning draws on complex networks of relationships: **causal models**
- linguistic causation: typically binary *cause-effect* relations

An alternative:

- causal language describes structures in online language-independent representations
- models can operate as ‘discourse parameters’ (providing relevant background for causatives and related predicates)
- discourse contributions interact (in familiar ways) with such representations; model relationships explicate linguistic effects
- causal models can also provide formal structure encapsulating generalized ‘world knowledge’ relevant for judgements about *causal* (but non-causative) language

(Nadathur & Lauer 2020, Baglini & Bar-Asher Siegal 2025, a.o.)

Telicity and culmination

Durative telic predicates (accomplishments) are associated with:

- **culmination (conditions):** object creation/destruction (*bake/eat a cookie*), terminus (*run a marathon*), state transition (*open a door*)
- **culmination assumption**: telic Ps *only denote culminated events*

Observation: we can refer to **non-culminated stages** of telic events

(1) Henny wrote a symphony. $\rightarrow$ *He completed it.*

a. Henny began to write a symphony (but gave up right away).

b. Henny stopped writing a symphony (and never began again).

Two questions:

1. **Analytical:** what governs truth, felicity of non/culminated uses?
2. What (**conceptually, lexically, semantically**) links relevant processes and culmination conditions?

Telicity and the imperfective paradox

Other contexts/devices for non-culmination:

- aspectual & related verbs (*begin, stop, end, try*)
- grammatical aspects: **progressives,** non-culminating perfectives (Hindi, Thai, many others; see, e.g., Martin 2019)
- aspectual(?) particles such as *frustratives* (e.g., Tohono O'odham, Copley & Harley 2014; Odam, Nadathur & Everdell 2025)

Our case study: the imperfective paradox (Dowty 1979)

- telic **progressives** are acceptable where culmination is precluded, clashing with the **culmination assumption**

(2) Henny was writing a symphony when she died. PAST+PROG
$\not\rightarrow$ *The symphony was eventually completed.*

Outline of the talk

[manuscript: ling.auf.net/lingbuzz/009209]

Outline of the talk

1. Introduction
2. The imperfective paradox
3. Expectation, culmination, and measurement: generalizing from the data
4. Causal models for telic predicates
5. Revisiting the data
6. Summary and outlook

[manuscript: ling.auf.net/lingbuzz/009209]

The imperfective (progressive) paradox

Telic **perfectives** often have **culmination entailments**:

(3) Maya wrote a book. $\rightarrow$ *A complete book came into being.*

Prevalent explanation:

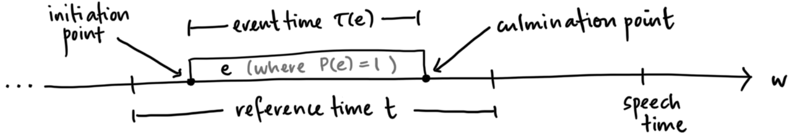
(i) **Culmination assumption:** $e \in [\![P_{\text{tel}}]\!]$ contains process + culmination

(ii) **Aspects instantiate** P**-eventualities** relative to reference time t

(4) $[\![\text{PFV}]\!] := \lambda w \lambda t \lambda P \,.\, \exists e[\tau(e) \subseteq t \wedge P(e)(w)]$

(cf. Klein 1994, Kratzer 1998, Bhatt & Pancheva 2005, a.o.)

Result: $e \in [\![P]\!]$ culminates, so t includes **culmination** ...



... leading to **culmination entailment**

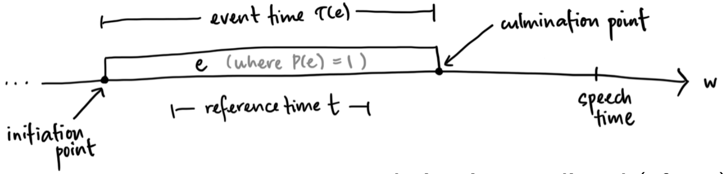
The imperfective (progressive) paradox

Wrong predictions for telic **progressives:**

If PROG instantiates $e \in [\![P]\!]$ as **ongoing** at t ...

$$[\![\text{PROG}]\!] := \lambda w \lambda t \lambda P \,.\, \exists e[\tau(e) \supseteq t \wedge P(e)(w)]$$

... **culmination assumption** requires culmination in w^*



... **culmination predicted** (after t)

Contradicts empirical data, leading to 'paradox':

(5) Henrietta was crossing the street (when she was hit by a truck).
no entailment: $\not\rightarrow$ *Henrietta reached the opposite side.*

Two assumptions, two approaches

Puzzle: why/when does PROG(P) apply to **partial P-eventualities**?
(pre-theoretical notion)

(A) **Intensional** PROG: **culmination** takes place in alternative worlds
(Dowty 1979, Landman 1992, Asher 1992, Bonomi 1997, a.o.)

- maintain **culmination assumption**, but allow PROG to **introduce modal alternatives**
- **analytical challenge:**
  constrain the modal relationship so that some P-**eventuality 'begins' in** w^*

Two assumptions, two approaches

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(pre-theoretical notion)

(A) **Intensional** PROG: **culmination** takes place in alternative worlds
(Dowty 1979, Landman 1992, Asher 1992, Bonomi 1997, a.o.)

(B) **Extensional** PROG: instantiate **non-culminated** P-eventualities
(e.g., Bach's 1986 'partitive puzzle', Parsons 1990, Szabó 2008)

- maintain **extensional** PROG, but **revise the culmination assumption**
- **analytical challenge:**
  what properties **qualify a partial ('process') eventuality** as *the same sort of thing as a culminated one* (i.e., what tells us that a non-culminated eventuality is something that aims at/makes progress towards culmination?)

Two assumptions, two approaches

Puzzle: why/when does PROG(P) apply to **partial P-eventualities**?
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(B) **Extensional** PROG: instantiate **non-culminated** P-eventualities
(e.g., Bach's 1986 'partitive puzzle', Parsons 1990, Szabó 2008)

Our claim: ultimately, we need both perspectives

- **evidently:** the notion of a part exists in relation to a (conceptual) whole
- **but:** part-whole relations belong to the concept of an *event type*, rather than the local context of initiation

Outline of the talk

Culmination, expectation on the intensional view

Intensional PROG: (Dowty 1979, Asher 1992, a.o.)
A P-eventuality is ongoing if reference time facts predict culmination

- PROG$(P) = 1$ iff events at t develop to culmination in **all normal/ inertial alternatives** to w^*

  Inertial alternatives $\sim$ ongoing processes continue uninterrupted

- alternatives are projected from circumscribed **perspectives** (Asher)

  (5) Henrietta was crossing the street (when the truck hit her).

  A set of facts including Henrietta, the truck, & their physical properties predicts collision as 'inertial' outcome

Upshot: truth of telic PROG is tied to likelihood, expectation (necessity) of culmination in relevant alternatives

Culmination, expectation on the intensional view

Intensional PROG:
$\text{PROG}(P) = 1$ iff ref time events develop into culminated P-eventualities in **all normal alternatives***

Impossible event (IE) progressives are out:

(6) **Context.** Meena's 5 year old daughter Maya wrongly believes that the earth is made entirely of sand and soil. She is digging a hole (with the intention of tunnelling all the way through).

Meena: **?**/✗ Maya is digging a hole to China.

- **false**: no (objective) perspective has normal culmination alternatives

*In Asher's terms: default/typicality inference is licensed (∼ completion occurs in the normal course of events based on state with the reference-time characteristics)

Culmination and expectation on the intensional view

Local culmination accessibility is a problem for **unlikely events (UEs):**

(7) Henrietta was crossing a minefield

(8) **The sailing competition** (*contra* 'defaults'; Bonomi 1997)
An international association organizes a sailing competition to circumnavigate the globe. 100 boats take part, and they all set sail from the same point. A few days later, a spokesman says:

✓ 100 boats are circumnavigating the globe. Most will fail.

- **intensional** PROG requires all normal continuations to culminate
  (or for each instance to license a default inference to completion)
- **so:** (7)-(8) are predicted to be **false**
- **because:** 'typical' attempts will not end successfully (failure is normal)

Empirically: (7)-(8) are both acceptable and **true**

Culmination, expectation on the intensional view

Solution? Capture **IE/UE contrast** by weakening intensional PROG to an existential

Dowty (1979) rejects this option based on an example attributed to Thomason:

(9) **The coin toss.** A fair coin is tossed and is still rising at utterance time.
  a. The coin is coming up heads.
  b. The coin is coming up tails.

- (9a) is true in half of the inertial futures, (9b) in the other half
- existential PROG makes both (9a), (9b) **true**, counter to intuition

(9) c. The coin is coming up heads or tails (but I don't know which)

Culmination, expectation on the intensional view

Existential intensional PROG is too strong for **out of reach (OOR)** contexts:

(10) **The un(der)trained runner** (cf. Szabó 2008, Varasdi 2014)
Benny began an ultramarathon for which he (knowingly) undertrained; it was certain before the start that he lacked the stamina to complete the run, but he meant go as far as he could.

a. *Friend/observer:* ✓Benny was running an ultramarathon (when he collapsed from exhaustion).
b. *Benny:* ✓I was running an ultramarathon (when I collapsed).

- predicted **false:** no situation containing Benny (+ relevant properties) is expected to continue to culmination

Empirically: (10)a-b are both acceptable and **true** in context

Similar example from Szabó (2004):

(i) As the architect was building the cathedral, he knew that, although he would be building it for another year or so, he couldn't possibly complete it.

The partitive perspective

Acceptability of **UE, OOR** progs suggests that telic PROGs don't need locally-accessible culmination alternatives, instead: "…it is the present activities that are the whole story." (Parsons 1989)

(11) **The part-of-proposal** (adapted from Landman 1992).
PROG(P) is true iff some actual event realizes **sufficiently much** of the type of events of P

The partitive concept implies a means of **measuring** the extent of a (partial) P-eventuality against some whole:

- a parallel between nominal and eventive *parts*: measurable features of the part allow comparison to the conceptual whole (Bach 1986, Parsons 1990)
- (*contra* Parsons) event partitivity cannot reduce to nominal partitivity: partial events can lack tangible correlates
- **intuition:** complex events involve organized/ordered stages ('steps' in a recipe), providing a conceptual standard of measure

Measuring accomplishments

Culmination procedures distinguish between **IE** vs. **UE, OOR** progressives:

Type	*Empirically*	*Intensional* PROG
IE	#	✗
UE	✓	✗
OOR	✓	✗

- **UEs, OOR** progs *can be true* if ref-time activities correspond to steps in a culmination 'recipe' for culmination (agent intentions can be evidence for a culmination procedure)
- **IE** progs are infelicitous: impossibility → no conceptual whole for comparison

Felicity, truth for telic progs (descriptive):

1. the existence of a realistic strategy/process for realizing culmination (per speaker's epistemic state)
2. actual events match (parts of) the culmination strategy, thus *making progress towards culmination*

Outline of the talk

Modeling progress: preview

Goal: combine intensionality, partitivity in the structure of telic predicates

- process-culmination relationship is **modal** in nature
  process leads to culmination in some *normative* sense
- actual events constitute **part of a** P**-eventuality**
  (but without local culmination expectation)

Idea: telic Ps invoke knowledge about necessary, sufficient conditions for initiating, developing, completing culmination processes

- a **type-level causal model** for culmination condition C_P provides 'recipe(s)' for realizing C_P
  + relevant preconditions (properties, facts, events) and relationships
- model induces a **(causal) mereological structure,** mediated through the relationship between process and C_P
- actual eventualities *partially realize* P if they conform to a causal pathway ($\sim$ normative/teleologically-optimal process) for C_P

Overview: structural equation models (Pearl 2000)

Causal information is represented by a **directed acyclic graph** D:

- **nodes** (finite set Σ): salient prop. variables (can be valued $u, 0, 1$)
- **edges:** atomic relations of **causal relevance** ($P \xrightarrow{\text{c-influences}} Q$)
- **structural equations:** specify how nodes' values are determined from their ancestors'

  Function Θ_D assigns to each $X \in \Sigma$ a pair $\langle Z_X, \theta_X \rangle$ where Z_X is the set X's immediate ancestors, $\theta_X : \{0,1\}^{|Z_X|} \to \{0,1\}$
- **causal consequences:** of a *situation* s (3-way valuation of Σ) are calculated using D and Θ_D

In lexical semantics:

Causal language refers to (predicates, presupposes) particular structural configurations as different causal dependency types

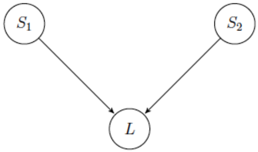
(cf. Nadathur & Lauer 2020, Baglini & Bar-Asher Siegal 2021)

Illustration: the Lifschitz circuit

(12) **The circuit example:** (Lifschitz 1990)

a. Suppose there is a circuit with two switches and one light, such that the light is on (L) exactly when both switches are in the same position (up or not up).

b. At the moment switch 1 is down, and switch 2 is up.



S_1	S_2	L
0	0	1
0	1	0
1	0	0
1	1	1

- (a) states the causal laws
- (b) gives us an initial setting (background situation)
- given (b), we expect the **causal consequences** to include that the light is off ($L = 0$)

Causal relationships in a model

Model structure allows us to define different causal relations:

(Nadathur & Lauer 2020, Baglini & Bar-Asher Siegal 2021)

- **background:** causation is a property of sets; effects are realized as the result of collections of conditions acting together
- causative predicates pick out causes with particular (binary) relationships to an effect *within a set of causes acting together*

Relations of interest (informally): (cf. Mackie 1965)

- **Causal necessity:**
  Within a situation s with both C and E, fact C is causally necessary for fact E iff changing C changes E
- **Causal sufficiency (of sets):** $\textsc{suff}^M(s, E)$
  A set s is sufficient for E iff E is true in s and s otherwise comprises necessary causes for E.

Causal necessity and causal sufficiency, formally

Let $M = \langle D, \Theta_D \rangle$ be a causal model over Σ, s a situation (3-way valuation of Σ)

(13) a. **Causal ancestors:** For $X \in \Sigma$, the **causal ancestors** of X are given by $A_X = \{Y \in \Sigma | R^T_{\Theta_D}(X, Y)\}$ (w/ $R^T_{\Theta_D}$ the trans. closure of direct ancestry)

b. **Facts.** A fact is a tuple $\langle X, x \rangle$ where $X \in \Sigma$ and $x \in \{0, 1\}$.

c. **Domain.** For s a situation, let $\textsc{dom}(s) = \{X \in \Sigma | \langle X, 1 \rangle \in s \vee \langle X, 0 \rangle \in s\}$

d. **Kernel.** For s a situation, let $\textsc{ker}(s) = \{\langle X, s(X) \rangle | X \in \textsc{dom}(s)\}$

e. **Consistency.** Situation s is **causally consistent** iff, $\forall X \in \textsc{dom}(s)$, $s(X) = f_X(s(Z_X))$

(14) **Causal necessity.**
Given a consistent situation s, $\langle X, x \rangle \in \textsc{ker}(s)$ is **causally necessary** in s for $\langle Y, y \rangle \in \textsc{ker}(s)$ iff $X \in A_Y$ and for any consistent s' s.t. $\textsc{dom}(s) = \textsc{dom}(s')$, $s(X) \neq s'(X) \rightarrow s - s' \supseteq \{\langle X, x \rangle, \langle Y, y \rangle\}$

(15) **Causal sufficiency.**
A consistent situation s is **sufficient** for a fact $\langle Y, y \rangle \in \textsc{ker}(s)$ iff:
$\forall X \neq Y \in \textsc{dom}(s), \langle X, s(X) \rangle$ is causally necessary for $\langle Y, y \rangle$ in s.
We call $S = \textsc{ker}(s)$ a **sufficient set** for $\langle Y, y \rangle$ $\qquad \textsc{suff}^M(S, \langle Y, y \rangle)$

Causal models: from the specific to the general

Past work: (Bar-Asher Siegal & Boneh 2020, Nadathur & Lauer 2020, a.o.)

- **statements of singular causation** (causative claims): about **token instances** of causation (actual relations between specific events)
- **observation:** singular causal claims depend in part on unobservable information

(16) Ria broke the vase.
  a. *Ria did something* (e_1)
  b. *The vase broke* (e_2)
  c. e_1 *(is part of what) brought about* e_2

- (16c): Ria is the agent/causer of some event which stands in a **recognized causal relationship** to a state transition involving the salient vase
- (16) is licensed by a **type-level** model of certain local relationships; its truth depends on actualization of e_1, e_2 and their temporospatial configuration w.r.t. the causal generalizations in a contextually-relevant model

Causal models: from the specific to the general

Telic predicates introduce type-causal information:

- accomplishments aren't **causative** (can hold where actual causation fails)
- but they can license singular causal claims (cf. Szabó 2004)

(17) Ria baked a cake

a. Ria engaged in some activity/series thereof (e_1) (✓$\textsc{prog}(P)$)

b. A cake came into being (e_2)

c. *e_1 is part of what brought about e_2*

Our claim:

- the truth of a telic perfective is not constituted by the causal relationship between a (prior) progressive and the associated result
- **but:** the predicate must provide information which *can* validate an actual causal relation between (prior) progressive and result (where both are realized)
- *ergo:* telic predicates provide the **same kind of information** that licenses token causal claims

Motivating causal complexity in event types

Claim: type-level models underlie lexicalization, use of complex predicates

- models capture generalizations/idealizations (causal regularities): *how things work* and/or *how to do things*
- type models support specific (token) expectations, but are not *about* singular instances (actual causation)

Accomplishments (durative telic predicates): **today**

- *bake a cake* $\sim$ perform a series of actions which, taken together, bring a cake into being
- felicitous use *presupposes* model; truth depends on match between model-actuality match
- type vs. token: possible to engage in an appropriate process *without realizing the type-level result* . . . resulting (for instance) in imperfective paradox effects

NB: The model is *not* a linguistic object (it has a language-independent existence), but it is the object which constitutes and supplies the "real" facts for linguistic, truth-conditional evaluation

Imperfective paradox: the view from causal models

Causal models: a framework for modeling progress that combines intensional, partitive perspectives on paradox effects

An **accomplishment event type** is a causal model M_P for predicate P:

- culmination condition C_P occurs in M_P as a dependent variable
- M_P links certain conditions/steps to one another and to C_P
- a **process** for P (a **causal pathway** S for C_P) is a set of jointly sufficient conditions for C_P $(S = \text{KER}(s)$, where $\text{SUFF}^{M_P}(s, \langle C_P, 1\rangle))$
- the model also specifies sufficient sets for **non-culmination** $(\langle C_P, 0\rangle)$

Beyond progressives: M_P induces a **mereological structure** where $[\![P]\!]$ contains (non-)culminated eventualities; e_1, e_2 are comparable as subsets if they belong to the same causal pathway for C_P

Analytic truth conditions for $\text{PROG}(P_{\text{tel}})$

Informally:
Given a model M_P for telic P with culmination condition C_P, the progressive is true at time t iff the situation s at t is a **possible cross-section of a non-culminated P-eventuality:**

(a) s realizes some part (condition Q) of a causal pathway for $\langle C_P, 1\rangle$ (initiation)

(b) s does not realize a complete pathway for $\langle C_P, 1\rangle$ (non-maximal)

(c) s does not realize a suff. set for non-culmination ($\langle C_P, 0\rangle$) (non-terminated)

Formally:
For telic predicate P with culmination condition C_P:

(18) $\text{PROG}(P)$ is true at world w, time t iff

$$\exists s \supseteq w[\tau(s) \circ t \wedge [\exists Q \exists S : Q \in S \wedge \text{SUFF}^{M_P}(S, \langle C_P, 1\rangle) \wedge Q(s) = 1] \quad \text{(a)}$$

$$\wedge [(\forall S : \text{SUFF}^{M_P}(S, \langle C_P, 1\rangle)[\exists Q \in S : Q(s) = 1 \rightarrow \exists Q' \in S : Q'(s) \neq 1]] \quad \text{(b)}$$

$$\wedge [\forall S : \text{SUFF}^{M_P}(S, \langle C_P, 0\rangle)[\exists Q \in S : Q(s) \neq 1]]] \quad \text{(c)}$$

Towards a compositional picture

Division of labour:

- The model provides conditions of **initiation**, **culmination**, **termination** (in terms of part-whole structure of sufficient sets)
- A **partitive** approach to grammatical aspects (Filip & Rothstein 2005, Altshuler 2014, a.o.): aspectual operator evaluates a(n actual) situation w.r.t. the appropriate event type model (how much / what sort of piece is instantiated)

Tentatively:

(19) $[\![\text{PROG}]\!] := \lambda w \lambda t \lambda P.\exists s \supseteq w[\tau(s) \circ t \wedge \texttt{INIT}(s, M_P) \wedge \neg\texttt{END}(s, M_P)]$
where

a. $\texttt{INIT}(s, M_P) := \exists Q \exists S : Q \in S \wedge \text{SUFF}^{M_P}(S, \langle C_P, 1\rangle) \wedge Q(s) = 1$

b. $\texttt{END}(s, M_P) = \texttt{CUL}(s, M_P) \vee \texttt{TERM}(s, M_P)$

i. $\texttt{CUL}(s, M_P) = \exists S : \text{SUFF}^{M_P}(S, \langle C_P, 1\rangle) \wedge \forall Q \in S, Q(s) = 1$

ii. $\texttt{TERM}(s, M_P) = \exists S : \text{SUFF}^{M_P}(S, \langle C_P, 0\rangle) \wedge \forall Q \in S, Q(s) = 1$

Sidebar: two kinds of perfectives within this approach

(20) **Hindi terminating (weak) perfective**

a. Maayaa-ne biskuT-ko khaa-yaa, par use puuraa nahiin
Maya-ERG cookie-ACC eat-PFV$_1$, but it.ACC whole not
khaa-yaa / #aur use ab-tak khaa rah-ii hai.
eat-PFV$_1$ / #and it.ACC now-until eat PROG-F PRES
'Maya ate the cookie, but did not finish it / #and she is still eating it.'

b. $\text{PFV}_1(P,t) := \exists s[\tau(s) \subseteq t \wedge \mathtt{INIT}(s, M_P) \wedge \mathtt{END}(s, M_P)]$

(21) **Hindi culminating (strong) perfective**

a. Maayaa-ne biskuT-ko khaa liyaa, #par use puuraa nahiin
Maya-ERG cookie-ACC eat PFV$_2$, #but it.ACC whole not
khaa-yaa.
eat-PFV$_1$
'Maya ate the cookie, #but did not finish it.'

b. $\text{PFV}_1(P,t) := \exists s[\tau(s) \subseteq t \wedge \mathtt{INIT}(s, M_P) \wedge \mathtt{CUL}(s, M_P)]$

(data from Singh 1998, Arunachalam & Kothari 2011; see also Nadathur & Filip 2021)

Outline of the talk

Culmination puzzles from a causal perspective

Big picture: causal models connect ✨world knowledge✨ to model/set-theoretic, truth conditional semantics

- this lets us write fairly intuitive truth conditions for (e.g.) progressives

Intensional PROG accounts cannot differentiate between **IE, UE**, and **OOR** PROGs, but the causal approach does:

1. **IE** PROGs are **infelicitous** (not false): event model does not exist
   - e.g., no set of conditions sufficient for digging a hole to China
2. **UEs, OORs** have models: truth thus depends on actual events
   - to complete an ultramarathon, one must show up at the start, take steps along the path, . . .
   - even though Benny's properties ensure eventual failure, PROG holds because his actions up to collapse match a culmination pathway
   - **upshot:** it's predictable that his endurance will fail, but Benny's actions until then can *make progress towards culmination*

Intentions, globally necessary conditions

Globally necessary conditions have a special status in the model: they must be sustained throughout the development of a P-eventuality

$$Q \text{ is a } \textbf{GNC} \text{ iff } \textsc{suff}^{M_P}(\{\neg Q, \langle C_P, 0\rangle\}, \langle C_P, 0\rangle)$$

Observation: Intentions act as GNCs for agentive accomplishments

(22) **Context.** Benny began running in a marathon at 9am. He sat down exhausted at 11:35, intending to end his run. But since he felt better after a short rest and refreshment, he decided to continue; he started running again at 11:50am.
Target: Benny was running a marathon

- **true** at 11:30, 11:55, **false** at 11:40

Captured by condition (c) for telic progressives:

(c) s does not realize a sufficient set for non-culmination $\neg\texttt{TERM}(s, M_P)$

See also Varasdi (2014) on sustaining vs. indicative conditions for telic Ps

Globally necessary conditions: a refinement

Q is a **GNC** iff $\textsc{suff}^{M_P}(\{\neg Q, \langle C_P, 0\rangle\}, \langle C_P, 0\rangle)$

GNCs are minimal conditions for *P*-eventualities to be in progress:

(cf. Bonomi 1997)

- no truth judgements where GNCs are underdetermined

(23) **Context.** Benny began running in a marathon (42K). Knowing he had undertrained, he intended to stop early. He planned to decide at 15K whether to stop there or continue to 21K. He collapsed at 10K, before making a decision. Later, he says:

Benny: **?** I was running a 15K/half marathon.

Presupposition: $\forall Q : \textsc{suff}^{M_P}(\{\neg Q, \langle C_P, 0\rangle\}, \langle C_P, 0\rangle), Q(s) \neq u$

Non-agentive GNCs: momentum, velocity (conserved quantities) for inanimate objects; see Bonomi (1997) for relevant examples

Outline of the talk

Summary

Imperfective paradox needs a combined intensional, mereological view:

- (locally-assessed) **culmination potential** cannot be the whole story
- the model provides a structure against which to measure the conditions under which a **token qualifies as partial realization (of type-defined whole)**
- **uninflected telic predicates** are 'responsible' for paradox effects (cf. Zucchi 1999, a.o.)

Progressives of accomplishments require causal knowledge but are **not themselves causal statements**:

- we need a (plausible) causal model to license $\text{PROG}(P)$
- use of $\text{PROG}(P)$ indirectly (via presupposition) conveys a speaker's belief in a causal model for P's culmination (a belief that there is a way to do P)
- **but:** asserted content only reports a match between actual events and the structure of the type-level model

Outlook: a new aspect calculus?

Dowty's (1979) aspect calculus: derive *Aktionsarten* from a decompositional analysis in terms of a small set of lexical atoms (`CAUSE`, `BECOME`, `DO`, `...`)

- **Causal models are the right kind of tool:** map world knowledge about (causal) interrelationships between 'conditions' to set-theoretic objects
- **Relata:** stative properties (manners, intentions, θ-roles), state transitions/point events
- `PROG` compatible with accomplishment Ps due to **structure/properties of model** (durativity, telicity)

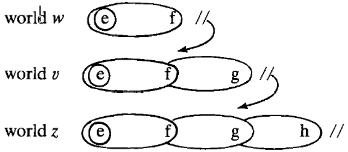
(Some of the) open questions:

- $[\![P]\!]$ as a **set of eventualities** vs. causal-model perspective
- **Aspectual composition:** how should *run* and *run a mile* be related?
- **Semantic roles** are plausibly represented as properties, but how does the assignment of individuals to roles occur?
- Causal models for **atelic** (and punctual) predicates?
  (no culmination conditions, but model might specify termination conditions instead)

Appendix: comparison with Landman (1992)

Landman: check for culmination in **continuation branch** of e in w

- building a continuation branch of event, world pairs:



Follow max development f of e in w, then move to closest v where f does not stop, iterate:

- e is a **stage of** $f, g, \ldots$
- $v, z, \cdots \in R(e, w)$

- **reasonable options:** $v \in R(e, w)$ iff there is a reasonable chance based on *what is internal to e in w* that e continues in w as far as in v

Like the causal approach:

- Landman's PROG centers the possibility of *continuation* or *continued development* of a partial P-eventuality at each modal transition
- relies on notions of *partial realization* (via *stage of* relation)

Comparison with Landman (1992)

Continuation branches capture the difference between **IE**, **UE** PROGs:

- CB construction removes interruptors in series, so **UE** PROGs are fine as long as each interruption could be 'reasonably' avoided

(6) Maya is digging a hole to China **NB:** ✗ for Landman

(7) Henrietta was crossing a minefield.

- **predicts** past-tense **IE** PROGs acceptable if task is unexpectedly completed: CB only looks at w^* continuations of e

(24) (I would never have believed it at the time, but) Maya was digging a hole to China!

On the causal approach:

- Maya's success is evidence of a legitimate causal pathway (i.e., whatever she did); the truth-conditional match follows
- **prediction:** difficult to apply in other scenarios (model lacks detail)

Comparison with Landman (1992)

Ultimately: Landman's PROG is intensional, existential ($\exists$ P-culmination in $R(e, w)$):

- does not explain **OOR** PROGs (locally no reasonable chance of completion)
- wrong preds for disjunctive/underdetermined cases (Bonomi 1997)

(25) **Context.** Meena was driving north from Monterey, intending to go to either SF or Oakland. At San Jose, she had not yet decided, when she got into a trip-ending accident.

a. ✓ Meena was driving to a Bay Area city (SF or Oakland)

b. **?** Meena was driving to SF

c. **?** Meena was driving to Oakland

- Landman: (25a) requires CB of w^* trip to culminate (in SF or Oak)
- **but:** neither of (25b,c) holds, not improved by overt uncertainty

(26) d. **?** Meena was driving to SF or she was driving to Oakland, but I don't know which

Comparison with Landman (1992)

The causal approach:

- **gives content** to the notions of *partial realization* (via causal structure), *stage of* (subset of a culmination pathway; condition a)
- **makes good on the intuition** that *continuation possibilities* matter more than culmination: condition b, condition c
- **actual events are normative parts** of culminating P-eventualities, culmination need not be locally *reasonable* (thus **UE, OOR** PROGs)
- un(der)determination: GNCs only met for undistributed disjunction
  - Meena intends trip to culminate in [SF $\vee$ Oakland] ✓(27a)
  - lacks specific intention for SF, Oakland **?**(27)b,c
- **NB:** epistemic uncertainty in non-agentive cases compare (24)d

(9) c. [while fair coin is in the air]
The coin is coming up heads or it is coming up tails, but I don't know which.

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